Self-Study: ChE 230 – Molecular Systems in Chemical Engineering – Spring 2026
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The strongest performance was observed for PI 2, with an average score of 2.35. This indicator assessed students' ability to solve engineering problems using algebraic relationships, specifically the derivation of the lever rule. Many students were able to follow the mathematical structure of the derivation and apply algebraic manipulation to relate phase fractions to alloy compositions. This suggests that students are developing reasonable confidence in using mathematical relationships once the variables and governing equations are clearly defined.

To improve student performance, future offerings of the course should include additional in-class examples that explicitly connect phase diagrams, microconstituents, and phase-rule calculations. More emphasis should be placed on helping students distinguish between phases and microconstituents, and on identifying which values from a phase diagram are needed for a specific calculation. In addition, student feedback indicated some perceived overlap between ChE 230 and another course in the curriculum. While some overlap is expected because of foundational topics such as phase behavior, thermodynamics, molecular structure, and material properties naturally support multiple chemical engineering courses, this feedback will be used to refine the course more broadly. Future offerings of ChE 230 will more clearly emphasize its distinct role as a molecular systems course by consistently connecting topics to molecular structure, bonding, microstructure, structure–property relationships, and engineering applications. Course content will also be reviewed to identify areas where coordination with related courses may reduce unnecessary repetition while preserving useful conceptual reinforcement.